# ACME Salary Management System

**Requirements Document — Version 1.0**

## 1. Goal

Build a secure, web-based salary management system that replaces Excel-based salary tracking for ACME's 10,000 employees across multiple countries. The system will enable HR managers to maintain employee and salary records efficiently and answer business questions about compensation through searchable data, reports, and analytics.

The primary objective is to establish a single source of truth for salary data, improve data accuracy, and make compensation insights accessible without manually consolidating spreadsheets.

## 2. Scope and Features

### A. Employee and Salary Management

- Maintain employee records, including employee ID, name, department, designation, country, and employment status.
- Create, view, update, and deactivate employee records.
- Maintain salary information, including salary amount, currency, and effective date.
- Preserve salary history when compensation changes rather than overwriting previous records.
- Support search, sorting, filtering, and pagination across employee records.

### B. Data Import and Validation

- Import existing employee and salary data from CSV/Excel files.
- Validate required fields, data types, currencies, duplicate employee IDs, and salary values.
- Display import errors with actionable feedback and prevent invalid records from being silently accepted.

### C. Salary Analytics and Reporting

- Provide summary metrics such as employee count, total salary expenditure, and average and median salary.
- Filter and group salary reports by country, department, and designation.

- Identify minimum and maximum salaries and highest-paid employees.
- Handle multiple currencies explicitly; avoid aggregating incomparable amounts without a defined conversion method.
- Export filtered reports for authorized users.

### D. Question Answering

- Allow HR managers to obtain answers to common compensation questions, such as average salary by country, salary expenditure by department, and employees above a specified salary threshold.
- Return results in clear tables or summaries, with relevant filters and calculation definitions.
- Initially support a defined set of questions using validated, parameterized database queries. Unsupported questions should receive a clear explanation rather than fabricated results.

### E. Security and Reliability

- Authenticate users and restrict access to authorized HR personnel.
- Record an audit trail of salary changes, including the user, timestamp, and relevant before-and-after values.
- Protect sensitive salary information and validate inputs consistently.
- Support at least 10,000 employee records with responsive, paginated searches and reports.

## 3. Deliberately Out of Scope

- **Payroll execution, payslips, tax deductions, and statutory filings:** These require country-specific rules and compliance expertise; they are not necessary to demonstrate salary data management and organizational insights.
- **Recruitment, attendance, leave, and performance management:** These are broader HRMS capabilities that do not directly address the stated problem.
- **Employee self-service and manager approval workflows:** The initial use case centers on HR managers; additional roles and approval processes can be introduced after validating the core workflow.
- **Fully open-ended AI analytics:** Natural-language-to-SQL generation introduces security, correctness, and data-leakage risks. A controlled question-answering layer is sufficient for the initial release.
- **Automatic foreign-exchange conversion and historical market-rate integration:** These introduce external dependencies and accounting assumptions. Initial reports will retain currency context and use an explicitly defined conversion policy if conversion is required.

- **Microservices, data warehouses, and advanced distributed infrastructure:** The expected scale of 10,000 employees does not justify this complexity for the initial release.

These exclusions keep the first version focused, testable, and deliverable while leaving room for future expansion based on demonstrated business needs.

# 4. Success Criteria

The initial release will be considered successful when HR managers can securely manage employee and salary records, import existing spreadsheets with clear validation results, retrieve accurate salary reports across supported filters, and answer the defined compensation questions without manually manipulating Excel files. Salary history and audit records must remain intact, and automated tests must verify core business rules, authorization, and reporting accuracy.

**Guiding principle:** Build a reliable salary data management and insights platform first; expand into a comprehensive payroll or HRMS product only when required by validated business needs.